Grey Is All Theory: Spatial Aggregation and the Limits of Civic Evidence in Leipzig's Open Data

Jonas Paul
Computational Spatial Humanities
Universität Leipzig
ab67ubeb@studserv.uni-leipzig.de

Lucas Berger
Computational Spatial Humanities
Universität Leipzig
al29luce@studserv.uni-leipzig.de

***Abstract*—Leipzig publishes 398 open datasets and satisfies every requirement open-data policy imposes. We ingested all of them into a single spatially and temporally joinable platform and measured what that took. Over nine weeks in which we ingested nothing the portal newly published, datasets carrying usable data rose from 152 to 350 and statistical records from 5,925 to 377,083; the difference is interpretive labour, delivered as five hand-maintained registries and several hundred curated indicator definitions. We report a taxonomy of the frictions responsible, including two that fail silently: 56 of 300 spatial metric names occur in more than one dataset with different base populations, and files mix administrative levels without marking them, which misplaced 13,862 rows in our own ingestion. A case study over Leipzig's 63 districts pairs 2023 housing, rent and income indicators: average residential floor area rises strongly with household income, while reported rents and floor area fall with welfare receipt. Recomputing one housing–social pairing at district scale changes the coefficient and the evidential force of the result. We argue that the publication is not transparency: the interpretive layer between them has a cost, and whoever can pay it decides what is transparent in practice.**

***Index Terms*—open government data, urban dashboards, spatial aggregation, modifiable areal unit problem, urban housing, critical data studies**

"Denn was man schwarz auf weiß besitzt, / Kann man getrost nach Hause tragen."[1]
— Goethe, Faust I, V. 1966 f.

I. Introduction

The city of Leipzig publishes 398 datasets from 37 municipal offices on an open data portal. By the standard any open-data policy applies, the city is transparent: the data are online, machine-readable, documented, and free.

We tried to use all of them. This paper reports what stood between the published files and a question anyone could answer with them, and it argues that the distance is not an implementation detail but the substance of the transparency claim. Publication and usability are separate achievements, purchased separately. Only the first is what open-data policy measures, and the second is what a citizen needs.

The platform we built to establish this is called *auerbachs-auge.tech*, after Auerbach's Cellar, the Leipzig tavern where Goethe set a scene of *Faust*, in which Mephistopheles bores a wooden table and pours wine that seems abundant until it is touched. It ingests both of Leipzig's data sources into a single spatially and temporally joinable model, serves them as vector tiles and time series, and computes correlations between any two indicators that share a spatial unit and a year.

So we contribute:

- A production system that brings an entire municipal open data catalogue (398 datasets from two sources with incompatible paradigms) into one model in which any two indicators can be compared, running on a single 4 GB server.
- An empirical taxonomy of the frictions separating published from usable data, each quantified against the full catalogue, including two classes that produce silently wrong results rather than visible failures: metric names that are not unique, and administrative levels mixed within a file without being marked.
- A worked spatial case study over Leipzig's 63 districts, reported together with the methodological guardrails it requires and with a demonstration, on the city's own housing and social indicators, that re-aggregating the analysis to 10 districts changes what a correlation appears to support.
- An argument, evidenced by our own constraints rather than asserted, that the interpretive layer between publication and use has a cost, that the cost is unevenly borne, and that whoever bears it decides what is transparent in practice.

The remainder of the paper follows that order: Section II situates the argument, Section III measures the frictions, Section IV describes the system built against them, Section V reports coverage and the case study, and Section VI takes up what the whole exercise implies for the civic promise of open data.

II. Background and Related Work

Open government data rests on a civic premise: publish the administration's numbers in machine-readable form and citizens gain the standing to check what they are told. European and German policy has largely settled the supply side of that premise, and municipal portals are now routine.

The critical literature has attacked the inference from supply to competence. Gurstein asks whether open data empowers anyone beyond those already equipped to use it, and argues

[1] "For what one owns in black and white, one can carry home with confidence."

that publication without effective use redistributes advantage rather than knowledge [1]. Kitchin, Lauriault and McArdle show that urban indicator systems and dashboards are not neutral windows but instruments that shape the governance reading them [2]. Mattern traces the dashboard's descent from mission control and its recurring promise of mastery through display [3]. Against the same tendency, Drucker argues that humanistic visualisation must treat its inputs as *capta* - taken and constructed rather than given [4].

Our methodological caveats come from spatial analysis. Tobler's observation that near things are more related than distant things [5] is why spatial correlation is worth computing at all. Openshaw's modifiable areal unit problem is why the result depends on the zoning chosen: aggregating the same observations into different areal units changes the coefficients, in both scale and shape [6]. Robinson's demonstration that ecological correlations do not transfer to individuals [7] bounds what any of it may be said to mean.

What is missing is the layer between. Leipzig's portal lists fifteen applications built on its data; each serves a single dataset or theme, and none joins across the catalogue. The critical literature describes what dashboards do to their readers, and the methodological literature describes how aggregation misleads, but neither measures what it costs to make a municipal catalogue comparable in the first place. That measurement is what we report.

III. The Data Landscape and Its Frictions

By the ordinary standards of open government data, Leipzig's portal is a good one. It runs on CKAN, exposes DCAT metadata, and publishes 398 datasets from 37 municipal offices across 13 thematic groups. Resources arrive in machine-readable formats, some are refreshed daily, the catalogue reaches from the tree cadastre to housing, social and mobility indicators, and the listed contacts answer. Nothing that follows is a complaint about a negligent administration. Every friction we describe is compatible with every open-data requirement we are aware of, and that is precisely the point: compliance is measured at the moment of publication, not at the moment of use.

We drew from two sources with incompatible paradigms. The CKAN catalogue at `opendata.leipzig.de` offers file resources whose format is whatever the publishing office exported. The statistical office at `statistik.leipzig.de` serves a JSON API in a wide-by-year layout, with years as columns and indicator codes as rows. Neither is wrong. They simply cannot be read by the same code.

Table I lists the eight classes of friction we encountered while bringing all 398 datasets into one queryable model, each with the extent we measured. Two deserve emphasis because they are the ones that produced silent errors rather than loud ones.

Metric names are not unique. A metric is identified in the published data by its name alone, but 56 of the 300 district-level metric names we loaded occur in more than one dataset. For example the name `Ausländer` ("foreign nationals") appears in six, denominated once against residents, once against registered residents, once against employees, once against the unemployed, once against people with a migration background, and once against welfare recipients. Six quantities, six base populations, one label. Nothing in the published metadata distinguishes them. The unit of identity is the pair (dataset, metric), and a consumer has to infer that.

Administrative levels are mixed without being marked. A single file routinely contains rows for districts, for the city districts that contain them, and for the city as a whole, with no column stating which level a row belongs to. The level has to be inferred from the area name, and the names do not support the inference: the city districts *Mitte*, *Südost* and *Nordwest* are each a substring of an unrelated district name *Grünau-Mitte*, *Zentrum-Südost* and *Zentrum-Nordwest*. Any name-similarity heuristic conflates them, and the affected rows then carry the population of a district under the code of a neighbourhood. In our own ingestion this misplaced 13,862 rows across 42 datasets; Section VI returns to what it took to notice.

The same file structure makes even the city's population ambiguous. Summed over the 63 districts, Leipzig had 632,560 inhabitants in 2024; summed over the 10 city districts, the same 632,560; the file's own city row reads 632,562. Two residents are carried without an district assignment and appear only in the total. Even the population of a city depends on which row you read.

IV. System Design

Section III fixes the requirement. If 398 datasets are to be compared rather than merely downloaded, they must share one spatial axis, one temporal axis, and one semantic registry. Everything in *auerbachs-auge.tech* exists to supply one of those three, and Fig. 1 shows the seven services that carry a dataset from the portal to a map in a browser.

A. Pipeline

Ingestion runs in four stages. Typed extractors fetch each resource over HTTP with retry and back-off, and record every attempt in a `raw_ingest` audit schema, so a dataset that silently stops updating is visible as a run that fetched nothing rather than as a chart that stopped moving. A `staging` schema holds intermediate normalisation, including the melt that turns wide-by-year statistical layouts into long records. Loaders then upsert into a normalised `core` schema, and materialised views in a `mart` schema are refreshed concurrently after each run. A nightly pass covers all datasets; a five-minute loop covers the three that are genuinely live.

B. The registries are the contribution

The part of this system that took the most work is not the pipeline. It is five hand-maintained JSON registries that supply the meaning the portal does not ship. `dataset_contracts.json` names, for each of the 398 datasets, which resource to fetch, in which format, on which schedule, and whether it carries geometry. `dataset_families.json` merges annual variants

TABLE I

EIGHT CLASSES OF FRICTION BETWEEN PUBLISHED AND USABLE DATA, WITH THE EXTENT MEASURED WHILE INGESTING ALL 398 DATASETS OF LEIPZIG'S OPEN DATA PORTAL. NONE OF THEM VIOLATES AN OPEN-DATA POLICY.

Friction	**Example**	**Extent measured**
Format heterogeneity	CSV with varying delimiters and encodings, GeoJSON, JSON, XLSX, SHP	portal-wide
Structural heterogeneity	wide-by-year layouts (indicator $\times$ year/quarter/school-year) require melting before they can be stored	174 datasets initially loaded 0 rows
Spatial-reference heterogeneity	district name, district number, electoral district, coordinates, or no spatial reference at all	no shared key across datasets
Temporal heterogeneity	update intervals from daily to never; static timetables tagged as live	16 marked live, 3 genuinely live
Semantic opacity	`Unnamed: 10` as a metric name; identifiers and spatial units stored as metrics; election years collapsed to 1994	1,351 "metrics" at city level
Entity fragmentation	one phenomenon split across annual datasets (given names 2014–2025, federal elections 2021 and 2025)	398 datasets overstate the number of distinct phenomena
Metric-name collision	`Ausländer` exists in six datasets with six different base populations	**56 of 300** district metric names occur in >1 dataset
Unmarked administrative levels	one file holds districts, city districts and the city total with no level column; the consumer must guess	**13,862 rows in 42 datasets** resolved to the wrong level

into one logical dataset with a year dimension, and carries per-dataset hints where a loader heuristic guesses wrong. `dataset_categories.json` reconciles the portal's CKAN groups with the statistical office's numeric category scheme. `election_definitions.json` maps result columns onto parties for each election, verified against the named vote shares the statistical API reports separately. `indicator_catalog.json` is a curated register of indicator names, units and topics; the user interface never exposes a raw metric name, because Section III is what raw metric names look like.

We report this plainly because it is the paper's central empirical claim about cost. These five files are not configuration in the ordinary sense. They are a few thousand lines of human judgement about what published numbers mean, written by two people over one semester, and they are the difference between 398 downloadable files and 398 comparable ones.

C. Resolving space

The single most consequential decision in the system is how a spatial reference becomes a key. Raw data name places inconsistently: `Zentrum-Südost`, `Zentrum-Suedost`, `02`, `Schönef.-Abtnaundorf`. A resolver maps each raw key through an alias table onto a canonical code in a boundary table holding Leipzig's 63 districts, 10 city districts and 819 electoral districts, trying exact alias, exact code and embedded code before falling back to fuzzy name matching. Every statistic that carries a spatial reference is joined on that canonical code, which is what makes a correlation between two unrelated datasets possible at all. It is also, as Section III noted and Section VI takes up, where our own worst error occurred.

D. Serving

Geometry is served as vector tiles generated in the database with PostGIS `ST_AsMVT` and cached per tile in Redis, invalidated by a version bump after each nightly run; 489,011 features across 35 datasets remain interactive on modest hardware because the client never receives the underlying geometry. Statistics are served as JSON by an asynchronous API. The client is a React single-page application drawing MapLibre GL vector tiles.

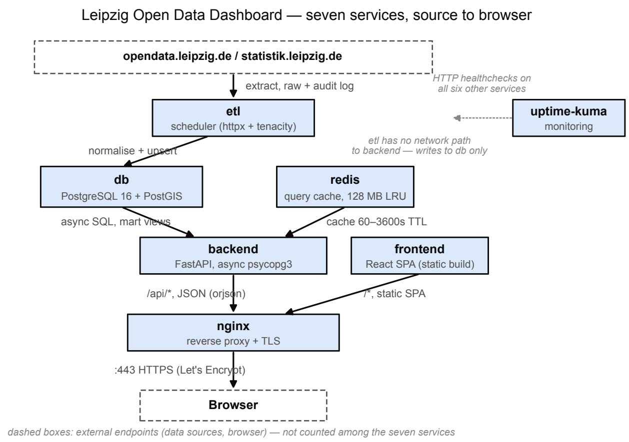

Fig. 1. The seven-service architecture of the dashboard and the path a dataset takes from the open data portal to the browser.

E. Operating envelope

The whole system, what means database, cache, API, ETL scheduler, static frontend, reverse proxy and monitoring runs in Docker Compose on a single virtual server with 4 GB of RAM, deployed from a Git. Access is authenticated. We return to these facts in the Section VI.

TABLE II

INGESTED CORPUS AT TWO POINTS NINE WEEKS APART, MEASURED AGAINST A REGISTRY FROZEN AT 398 ENTRIES. THE PORTAL PUBLISHED 15 FURTHER DATASETS IN THAT WINDOW; NONE OF THEM ENTERED THE PLATFORM.

	10 Jun 2026	**15 Aug 2026**
Datasets in our registry	398	398
carrying data	152 (38 %)	**350 (88 %)**
Statistical records	5,925	**377,083**
across datasets	122	**299**
Election result rows	0	**79,553**
Geo datasets	29	35
Geo features	4,175,146	489,011*
Traffic restrictions	30,999	1,421*

* lower after de-duplication; the smaller figure is the correct one

V. RESULTS

A. Coverage, and what the coverage measures

Table II reports the state of the ingested corpus at two points nine weeks apart. Datasets carrying usable data rose from 152 to 350 of 398; statistical records from 5,925 to 377,083; election results from none to 79,553.

Almost none of that came from the portal. It was not idle in those nine weeks: it created 15 datasets and modified 86. But our registry stayed frozen at the 398 entries of the first audit, so 13 of the 15 new datasets never entered the platform at all, and the two whose names we already carried are fetched from the statistical API rather than from the portal record. Of the 86 modified datasets, 69 were already registered with us, and 58 of those are likewise API-fetched, where a changed portal record cannot affect how they load. At most 11 datasets could have been helped by anything the city did in that window. The corpus grew by 198.

The remainder is interpretive labour: melt rules for wide-by-year layouts, alias entries for place names, column-to-party mappings, curated indicator definitions. That is what makes the table a measurement rather than a progress report. *The distance between published and usable data is the size of that delta.*

Two rows fall rather than rise, and both are corrections. Geometry features drop from 4,175,146 to 489,011 and traffic restrictions from 30,999 to 1,421 because several sources issue fresh feature identifiers on every fetch, so each nightly run re-inserted a full snapshot until the loaders learned to sweep what a run did not touch. The smaller numbers are the correct ones.

What survives as an analytical surface is narrower than 350 datasets suggests: 341,945 records across 67 datasets and 300 metrics carry a resolved district code, and five elections are stored at electoral-district level, four of them also aggregated to districts.

B. A worked case: housing, rent and income

We pair 2023 indicators from four small-area statistical datasets: reported apartment rents, household and personal net income, residential floor area, and SGB-II welfare receipt. The platform only compares observations with the same year and the same spatial unit, and it keys every metric by both dataset and name; the labels *Haushaltseinkommen* and *Gesamtmiete* are otherwise too little context to be safe.

The strongest relationship in this housing grid links average residential floor area per dwelling to household income ($r = +0.772$, $R^2 = 0.596$, $n = 63$). Larger dwellings are concentrated in higher-income districts, from $51.9\,\mathrm{m}^2$ per dwelling in Zentrum to $110.7\,\mathrm{m}^2$ in Baalsdorf. Rent indicators show the same social gradient, but with survey missingness made visible by the smaller sample: the reported total rent metric rises with personal income ($r = +0.677$, $R^2 = 0.458$, $n = 51$; Fig. 3) and falls with the SGB-II quote ($r = -0.667$, $n = 51$). The rent surface itself ranges from 8.08 in Grünau-Ost to 12.60 in Zentrum, with missing districts left uncoloured in Fig. 2.

Two further pairings show why the dashboard must report the denominator and the sample size next to the coefficient. Wohneigentum rises with household income ($r = +0.615$, $n = 62$), while rent burden falls only moderately with household income ($r = -0.386$, $n = 50$). A user who saw only the coefficients could mistake survey coverage, base populations and metric definitions for substantive urban structure. The point of the platform is to make the comparison possible without letting those caveats disappear.

C. The same data, zoned differently

Leipzig's 63 districts nest into 10 city districts, but not every published metric survives that move. The public API contains Stadtbezirk rows for rent, income and survey-based housing variables, yet those rows carry missing values in 2023; the platform therefore refuses a district-level rent–income correlation instead of silently manufacturing one. The housing–social pairing available at both levels is residential floor area per dwelling against the SGB-II quote: $r = -0.667$ over districts ($n = 63$) and $r = -0.758$ over city districts ($n = 10$; Fig. 4).

This is Openshaw's scale effect [6], and it is the reason this paper is named after Mephistopheles' warning to the student that all theory is grey [8]. Neither coefficient is the true one. At $n = 10$ a single district can carry the fit, the coarser unit suppresses within-district variation, and no aggregate correlation licenses any claim about individual households [7]. The finding is not that floor area causes income or welfare receipt. It is that a competent analyst, working honestly from published municipal data, must choose a zoning before the coefficient exists.

The same machinery supports other civic questions without changing the data model: election results can be compared with social or housing indicators; mobility layers with population density or air-quality measurements; tree, green-space and land-use data with heat, rents or traffic exposure; and public services such as schools or childcare with demographic change. These are not separate applications. They are different pairings on the same resolved spatial and temporal axes.

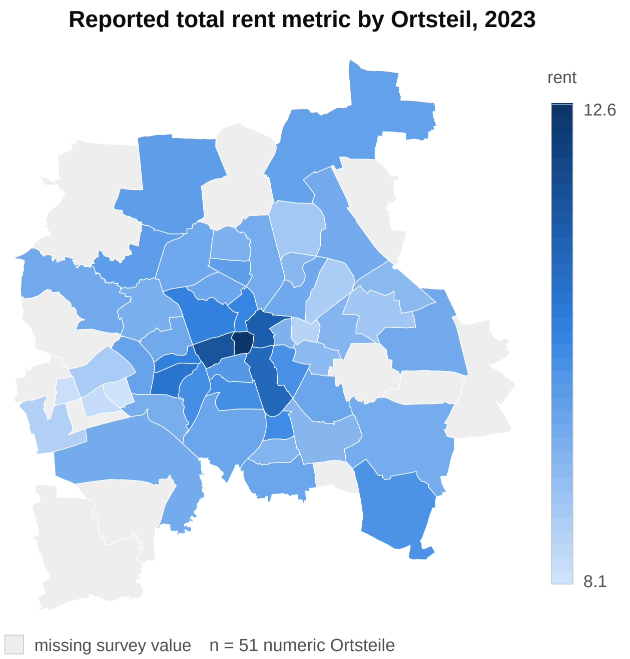

Fig. 2. Reported total rent metric per district in 2023. Survey missingness leaves 12 districts uncoloured; numeric values range from 8.08 in Grünau-Ost to 12.60 in Zentrum.

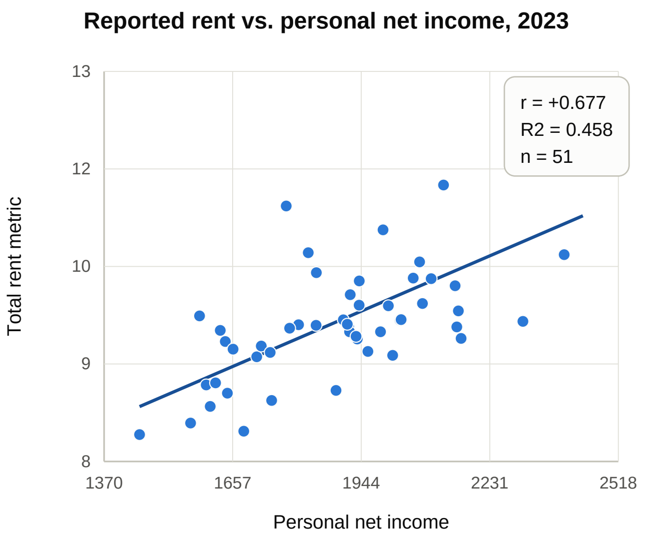

Fig. 3. Reported total rent metric against personal net income across 51 districts with numeric 2023 survey values ($r = +0.677$, $R^2 = 0.458$).

VI. Discussion: Transparency and Its Infrastructure

Open data is defended on the ground that it makes citizens competent to judge: given the numbers, people can form views from evidence rather than from assertion. We hold that position. This paper is an argument about what it costs to make it true.

A. *Publication is not the expensive part*

Leipzig discharged its obligation the moment the 398 datasets went online. What remained was the work in Section IV: five hand-maintained registries, roughly 300 metric definitions, an alias table for place names, and a resolver deciding what a row's spatial reference means. None of that is discoverable from the data. All of it is required before two datasets can be compared. Gurstein's objection to open data (that it empowers those already equipped [1]) names exactly this gap, and Table I is a measurement of it. A citizen with a spreadsheet and an afternoon does not clear it. Two computer science students with a semester barely did.

B. *What it took to notice our own error*

Section III described a source property: administrative levels mixed in one file with no column marking them. We were the consumer that guessed, and we guessed wrong. Our resolver's fuzzy fallback matched the Stadtbezirk *Mitte* onto the district *Grünau-Mitte*, and two of its siblings likewise, putting 13,862 rows across 42 datasets under the wrong spatial code and inflating Leipzig's 2024 population to 805,218.

Nothing about the resulting data looked wrong. Every map rendered, every chart had a plausible shape, every correlation was finite and interpretable. We found the error only because a total we computed disagreed with a total we already knew. That is worth stating precisely, because it defines the standard: data carrying the wrong meaning is not visibly different from data carrying the right one, and no amount of looking at a dashboard reveals the difference. Drucker's point that data are *capta* (taken, not given, constructed as an interpretation [4]) is not a philosophical caveat here. It is an operational one. Someone constructed those spatial codes, that someone was us, and we were wrong.

C. *Transparency has an infrastructure cost*

Our platform requires a login, and its metric list is curated rather than complete. Both cut against the openness we are arguing for, and neither is ideological. The system runs on a single 4 GB server: we have no budget for rate-limiting abuse or absorbing a crawl, and an uncurated metric list surfaces `Unnamed: 10` beside real indicators, which misleads more than it informs. They are capacity decisions.

That is the finding, not an apology. Two people who wanted maximal openness could not afford it. The interpretive layer that turns published data into usable data costs money, skill and continuous maintenance, and whoever can pay for it decides what is in practice transparent. Publishing more datasets does not distribute that capacity; it presumes it. If open data is to serve the competence it promises, the interpretive layer belongs to public infrastructure rather than to volunteers, and cities should be measured on whether their data can be used rather than on how much of it exists.

Four changes would do more for usability than any additional dataset: a stable spatial key shared across datasets; long-format publication instead of wide-by-year; machine-readable indicator definitions naming each metric's denominator; and

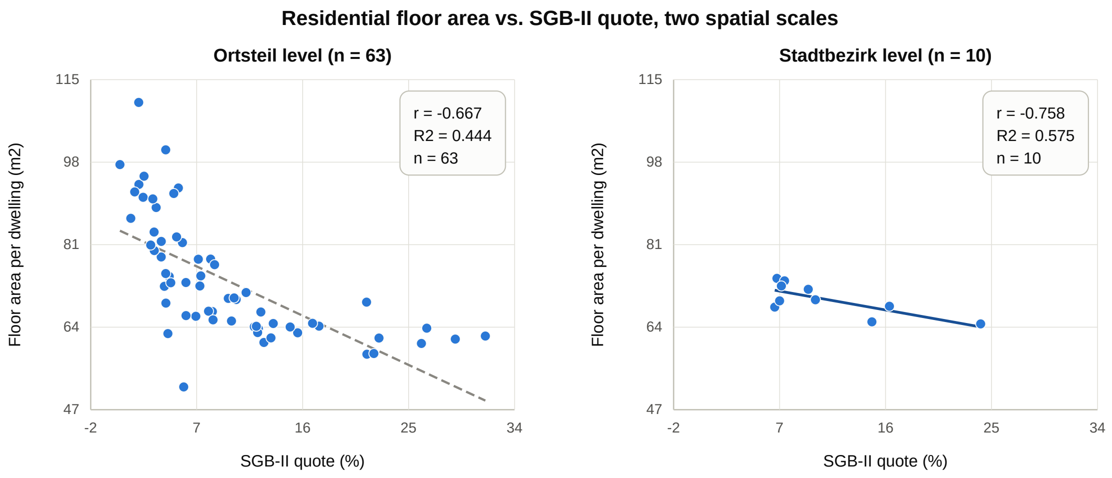

Fig. 4. Residential floor area per dwelling against the SGB-II quote at two spatial scales (district, $n = 63$, vs. Stadtbezirk, $n = 10$). Neither scale is the correct one, and the resulting $n = 10$ coefficient must not be read as a more authoritative finding.

one dataset per phenomenon with a year dimension instead of one per year.

D. What a dashboard cannot do

Mattern's history of the urban dashboard warns against the genre's implicit promise of mastery through display [3], and Kitchin et al. show how indicator systems shape the governance that reads them [2]. Our own results argue the same way from the inside. The platform will happily show a user the correlation in Section V-C at either spatial scale, and it cannot tell them which to believe, because the data cannot. A dashboard lowers one barrier: it makes comparison possible. It does not make comparison sound. Evidence-based citizenship needs both, and the second is a harder engineering problem than the first.

VII. Limitations and Future Work

Our platform is authenticated, so it does not itself demonstrate the public access it argues for. Forty-eight of the 398 datasets still carry no data, mostly statistical layouts our melt rules do not yet recognise. The case study covers one city and one reporting year; depending on the pairing, n ranges from 49 to 63 districts, and the district-scale comparison has only $n = 10$. Several rent and income variables are survey estimates with missing small-area values, and no correlation is a causal claim. The before-and-after coverage figures come from two audits of our own database rather than from an independently reproducible baseline; the portal-side counts beside them are reproducible, from the CKAN `package_search` API on `metadata_created` and `metadata_modified`. Accessibility of the interface is unaddressed. And the whole system depends on the portal continuing to publish in the shapes we learned.

The obvious extension is a second city, which would separate what is true of municipal open data from what is true of Leipzig. Beyond that, the registries in Section IV are the artefact most worth sharing: if a comparable catalogue can be described in the same form, the interpretive layer becomes portable rather than rebuilt each time.

VIII. Conclusion

Leipzig publishes 398 datasets and satisfies every open-data requirement we know of. Making them comparable took five hand-maintained registries, several hundred curated indicator definitions, and a spatial resolver whose failure we did not detect until a computed population disagreed with a known one. Between those two states we took nothing new from the portal; the labour did the work. That labour is the real subject of the transparency claim. Our case study finds Leipzig's housing surface stratified by income and welfare receipt, and finds the same housing–social relationship changing strength when the city is divided into larger districts. Both findings are available to any citizen who downloads the data. Neither is available to a citizen who only downloads it.

References

[1] M. B. Gurstein, "Open data: Empowering the empowered or effective data use for everyone?" *First Monday*, vol. 16, no. 2, 2011.

[2] R. Kitchin, T. P. Lauriault, and G. McArdle, "Knowing and governing cities through urban indicators, city benchmarking and real-time dashboards," *Regional Studies, Regional Science*, vol. 2, no. 1, pp. 6–28, 2015.

[3] S. Mattern, "Mission control: A history of the urban dashboard," *Places Journal*, Mar. 2015.

[4] J. Drucker, "Humanities approaches to graphical display," *Digital Humanities Quarterly*, vol. 5, no. 1, 2011.

[5] W. R. Tobler, "A computer movie simulating urban growth in the detroit region," *Economic Geography*, vol. 46, pp. 234–240, 1970.

[6] S. Openshaw, *The Modifiable Areal Unit Problem*, ser. CATMOG. Norwich: Geo Books, 1984, no. 38.

[7] W. S. Robinson, "Ecological correlations and the behavior of individuals," *American Sociological Review*, vol. 15, no. 3, pp. 351–357, 1950.

[8] J. W. Goethe, *Faust. Der Tragödie erster Teil*, 1808.